SiteProof

MATERIAL DELIVERY & SUPPLIER MGMT

SiteProof:

Smart Delivery & Supplier Management Platform

Product Requirements Document & Business Plan

Team Members

Po-Hao Cheng M11505503 — Software Engineer / System Architect

A. Samuel Arzamendia M11405807 — Business Analyst & Quality Control

1. Executive Summary

SiteProof is a web-based supply chain management platform specifically designed for General Contractors (GCs) and Material Suppliers (MS) in Taiwan.

Unlike traditional, heavy, expensive, and difficult-to-implement construction ERP systems, SiteProof intentionally chooses NOT to replace an ERP. Instead, it focuses sharply on one critical site workflow: the Material Delivery Closed-Loop. The platform unifies the scheduled delivery, supplier delivery status, physical site arrival, quality document verification, and retention money calculation into One Single Delivery Record. Through a lightweight architecture, SiteProof aims to help general contractors control delivery schedules and costs and reduce jobsite downtime (to be validated in a pilot), while helping well-performing suppliers build a digital performance record.

2. Problem & Value Proposition

2.1 Three Core Management Pain Points in Taiwan's Construction Industry

- **Delivery Visibility (Schedule & Staging Bottlenecks):** Urban jobsites in Taiwan have highly constrained physical space. Materials failing to arrive via Just-In-Time (JIT) delivery lead to site congestion, while delays cause worker idle time and severe cost overruns.
- **Receiving & Quality Traceability:** Quality inspection reports (e.g., 7-day and 28-day concrete compressive strength test reports) and ESG green building certifications are mostly paper-based, making field and back-office verification extremely time-consuming.
- **Cost & Supplier Accountability:** Budget volatility risks and the lack of a unified digital record make valuation and performance tracking inefficient. Cross-referencing paper delivery slips during monthly valuation is cumbersome, and there is a lack of systematic performance tracking after project completion.

2.2 Value Proposition

For General Contractors (GC)

- **Budget & Schedule Control:** SiteProof aims to reduce procurement-related inefficiencies and material-related waiting time; the magnitude of these improvements will be validated during pilot projects.
- **Quality & ESG Compliance:** Digitalization of CNS/ISO quality inspection reports and green/carbon labels makes alignment with government and owner audits easier to manage.
- **Objective Supplier Vetting:** A post-completion rating and delivery history database supports supplier performance tracking and preferred-supplier lists.

For Material Suppliers (MS)

- **Streamlined Delivery & Compliance:** Suppliers update delivery status and upload the required quality documents directly to the delivery record. Digital valuation payment requests follow in Phase 2.
- **Building Digital Credit Assets:** On-time delivery and complete quality records build a visible performance history that can help suppliers secure future contracts with general contractors.

3. Business & Revenue Model

3.1 Core Differentiator: One Single Delivery Record

SiteProof is intentionally designed around the practical realities of construction sites, avoiding complex data entry or extensive training. By making the Delivery the central record, the platform connects the delivery schedule, quantities, quality documentation, acceptance result, retention and supplier rating in one place. Contractors improve visibility and accountability without the complexity and cost of a full ERP implementation.

Scope note. Purchase orders are agreed outside SiteProof (in the contractor's existing process or ERP). SiteProof does not issue or negotiate orders; the contractor registers each agreed order as a Delivery Record and the platform tracks it from scheduling to completion.

3.2 Revenue Model

To align with the tight profit margins of the construction industry and the massive transaction volumes of bulk materials, SiteProof strictly charges 0% transaction fees, ensuring material procurement costs are not inflated for contractors. Revenue relies exclusively on two streams:

- **SaaS Subscription Tiers:**
  - Free Tier: single-project management and limited supplier connections.
  - Pro / Enterprise Tier (monthly/annual subscription): tailored for medium-to-large contractors, offering multi-project / multi-office management, advanced analytics, unlimited supplier connections, and ESG carbon footprint tracking. These Pro-tier features are delivered from Phase 2 onward; the Phase 1 MVP is built for the pilot and does not include them.
- **Featured Supplier Listing:** Qualified suppliers can pay for priority placement in search results for popular materials (clearly marked as sponsored, without altering their actual performance ratings). Introduced in Phase 2; it is not part of the MVP.

3.3 Customer Acquisition

- **Ecosystem Expansion via Top-Tier GCs:** Partner with 1–2 mid-to-large general contractors for initial pilots. Register their existing Qualified Supplier Lists (QSL) on the platform, bringing their material suppliers onto SiteProof in the process, then expand to small and mid-sized GCs.
- **Regulatory & ESG Alignment:** Highlight support for domestic Green Building Certifications and Net-Zero Carbon Emissions policies to attract ESG-focused listed contractors.

3.4 Competitive Edge

- **Lightweight & Complementary:** Compared to massive, expensive legacy ERP systems, SiteProof focuses specifically on the "Material Procurement & On-Site Ingress Closed Loop", designed for deployment within weeks (to be confirmed in the pilot) and for coexisting with the systems contractors already use.
- **Localized Engineering Logic:** Built directly around Taiwan's construction practices: ready-mix concrete plant delivery certificates, 28-day sample test reports kept on the delivery record (reminders arrive with LINE notifications in Phase 2), and automated retention calculation at the contract rate (configurable per contract, e.g., 3%–10%).
- **Fits Existing Habits:** Site crews and legacy suppliers already coordinate through LINE and phone calls. Future LINE integration (Phase 2) will push delivery status and time-slot

confirmations as LINE notifications, so adoption does not require abandoning current workflows.

4. SWOT Analysis & Cross Strategies

	Strengths (S)	Weaknesses (W)
Internal	• Focused coverage of the three core delivery pain points • Custom-designed for Taiwan construction workflows (JIT, valuation, sample testing, retention) • Built-in supplier performance ratings and delivery history	• Insufficient initial user base on both sides of the platform (cold-start problem) • Limited API integration with legacy ERP systems • Team development resources concentrated on the MVP core modules

	Opportunities (O)	Threats (T)
External	• Digital transformation trends in Taiwan's construction industry and BIM integration • Net-Zero Carbon targets driving demand for green building materials • Volatility in bulk material prices driving demand for real-time tracking	• On-site engineers and legacy suppliers are accustomed to using LINE or phone calls • Underperforming suppliers may resist transparent performance evaluations • Major ERP vendors may introduce competing modules in the future

Cross Strategies

- **SO Strategy:** Use the low-barrier web interface to expand to small and mid-sized general contractors in urgent need of digital transformation.
- **WO Strategy:** Offer free onboarding incentives to invite large contractors to bring their existing supply chains onto the platform.
- **ST Strategy:** Turn objective supplier ratings into a visible performance history that suppliers can use to win new orders.
- **WT Strategy:** Focus the MVP stage strictly on the "Delivery Tracking Closed Loop" without blindly developing bloated features.

5. Requirement Analysis

5.1 User Stories

Each story follows the course template: As a <role>, I want <feature>, so that <benefit>.

ID	Role (<role>)	Desired feature (<feature>)	Purpose & value (<benefit>)
US1	Contractor / Project Manager, Material Supplier, Site Inspector	I want to sign in with a role-based account,	so that I only see and do what my role allows.
US2	Contractor / Project Manager	I want to create a delivery record containing material details, supplier, value, and scheduled time slot,	so that I can establish a single source of truth for each material order.
US3	Contractor / Project Manager	I want to view delivery status, history, the calculated retention and supplier ratings,	so that I can streamline the valuation and payment process.

ID	Role (<role>)	Desired feature (<feature>)	Purpose & value (<benefit>)
US4	Contractor / Project Manager	I want to complete a delivery and rate the supplier (1–5),	so that our company can build an objective performance database.
US5	Material Supplier	I want to view assigned deliveries and update their status (Preparing → In Transit → Delivered),	so that the contractor knows exactly when materials will arrive.
US6	Material Supplier	I want to upload required delivery and quality documents (either at delivery or when available),	so that I comply with quality and ESG requirements without paper trails.
US7	Material Supplier	I want to view my performance rating and delivery history,	so that I can leverage my good record to win future contracts.
US8	Site Inspector	I want to record the actual quantity received and review the uploaded PDF/JPG documents,	so that I can verify the physical delivery matches the record.
US9	Site Inspector	I want to explicitly accept, partially accept, or reject a delivery,	so that quality control is strictly enforced before finalizing the delivery.

5.2 System Use-Case Diagram

Every user story maps to one use case (US1 → UC1, and so on). UC10 is the system behavior that the acceptance decision (UC9) always includes.

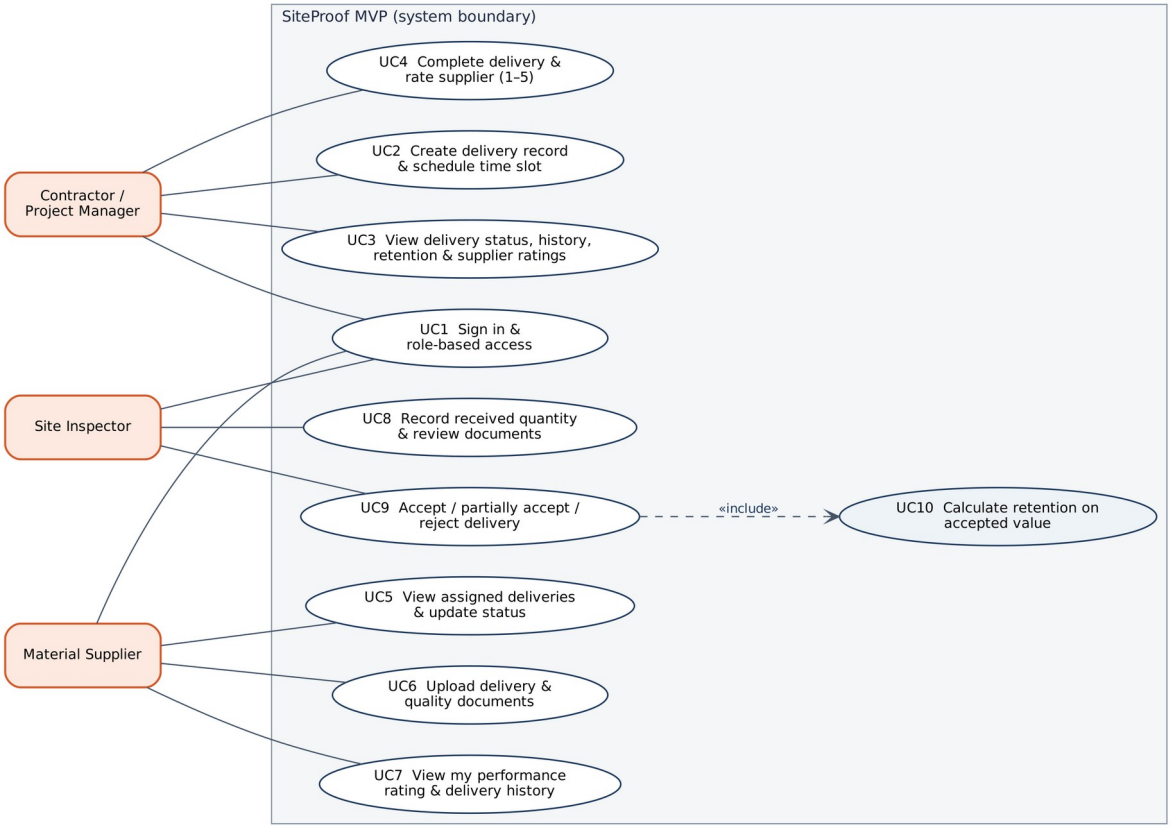


Figure 1. SiteProof MVP use-case diagram (Mermaid-style notation; UC10 is included by UC9).

5.3 Business Sequence Diagram

The flow starts directly with the creation of the delivery record (no RFQ / quotation phase in the MVP) and includes the three acceptance outcomes before completion.

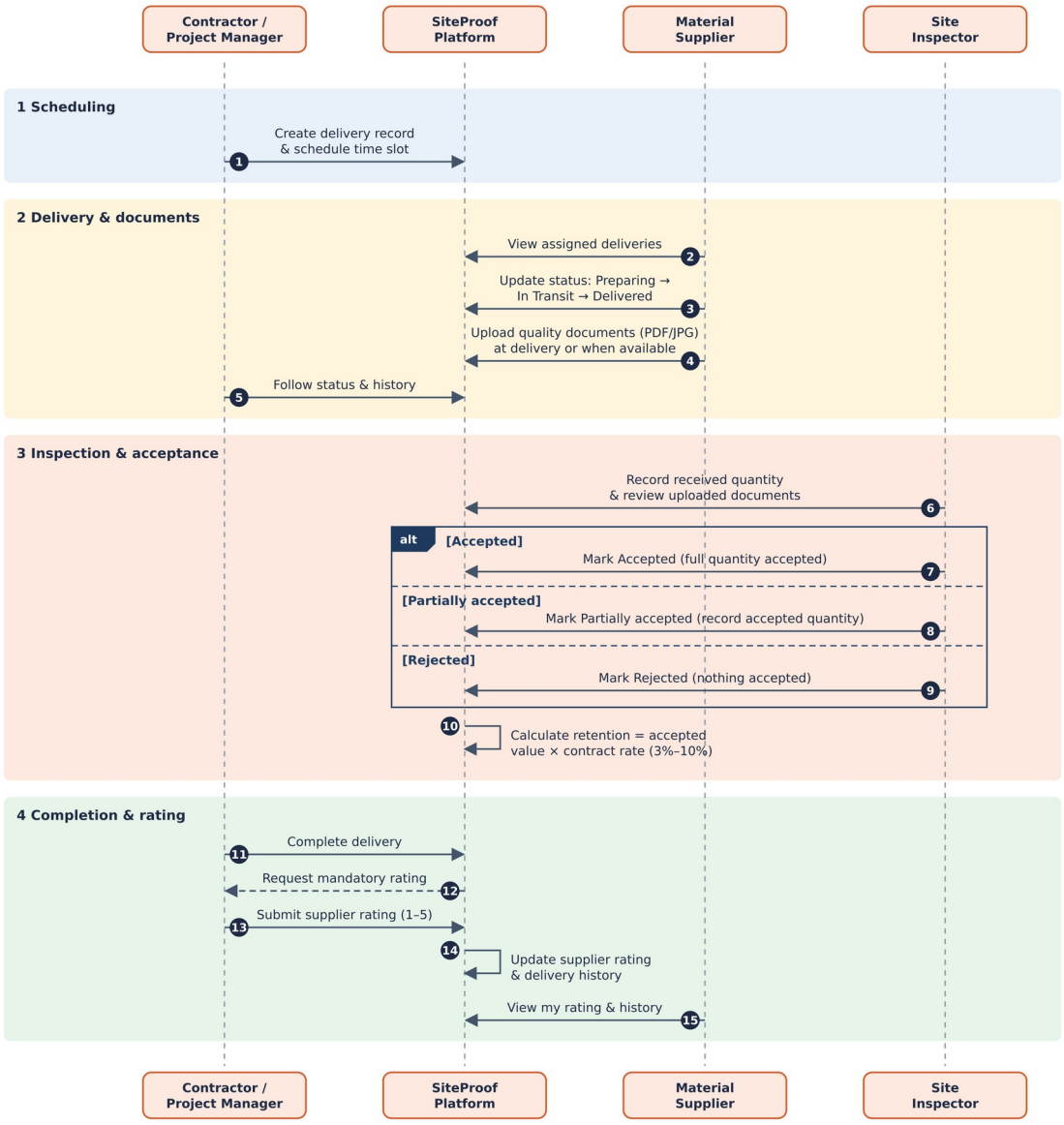


Figure 2. Business sequence diagram of the Material Delivery Closed-Loop.

5.4 Key Business Rules

- **Delivery lifecycle:** Scheduled (on creation) → Preparing → In Transit → Delivered → inspected (Accepted / Partially accepted / Rejected) → Completed.
- **Acceptance outcomes:** Accepted = all the quantity received is accepted. Partially accepted = the inspector records the accepted quantity, which is less than the quantity received. Rejected = nothing is accepted. In every case, accepted value = delivery value × accepted quantity ÷ ordered quantity on the delivery record, and the accepted quantity cannot exceed the ordered quantity.
- **Retention:** Retention = accepted value × the retention rate configured in the project's contract (e.g., 3%–10%). It is calculated automatically when the inspector records the acceptance decision (UC10); a rejected delivery has no accepted value and therefore no retention.

- **Completion and rating:** After the acceptance decision, the project manager completes the delivery and must give a 1–5 supplier rating, whatever the outcome. Rectification or re-delivery workflows are outside the MVP.
- **Documents:** Quality documents (PDF/JPG) can be uploaded at delivery or later, when they become available (for example, 28-day test reports). The inspector reviews the documents uploaded at the time of inspection.
- **Notifications:** The MVP has no push notifications; users see delivery status on the platform. LINE notifications arrive in Phase 2.

5.5 Traceability: MVP ↔ User Stories ↔ Use Cases

MVP must-have	User stories	Use cases
1. Account & Role Management	US1	UC1
2. Delivery Management	US2, US3	UC2, UC3
3. Delivery Status Tracking	US5	UC5
4. Receiving & Quality Documentation	US6, US8, US9	UC6, UC8, UC9
5. Retention & Completion	US4, US7	UC4, UC7, UC10

6. Minimum Viable Product (MVP Definition)

Applying the MoSCoW method, the core feature set required for SiteProof's Phase 1 market launch includes:

Must Have (MVP Core)

1. **Account & Role Management:** Role-based access for Contractor, Supplier and Site Inspector.
2. **Delivery Management:** The contractor creates delivery records containing material, specification, quantity, supplier, delivery value and scheduled time slot, and can view their status and history.
3. **Delivery Status Tracking:** The supplier updates the status: Preparing → In Transit → Delivered.
4. **Receiving & Quality Documentation:** The inspector records the actual quantity received, reviews the uploaded PDF/JPG documents (either at delivery or when the document becomes available), and marks the delivery Accepted / Partially accepted / Rejected.
5. **Retention & Completion:** The system calculates retention on the accepted value at the project's contract rate (e.g., 3%–10%). The project manager then completes the delivery and provides a mandatory 1–5 supplier rating.

Should Have — Phase 2

RFQ & quotation comparison, LINE notifications (including 28-day test reminders), digital payment requests, advanced analytics & Pro-tier features, featured supplier listings, mobile optimization.

Could Have — Phase 3

OCR for quality inspection PDFs, budget-overrun alerts, GPS / logistics integration, ERP integration, predictive delivery-risk analytics.

7. Team & Validation

Team

- **Po-Hao Cheng:** Software design and development, data modeling, and platform architecture.
- **A. Samuel Arzamendia:** Construction-site knowledge, procurement and delivery workflows, and the voice of the general contractor.

Validation to Date

The three pain points and the MVP are built on field experience as a civil engineer, research on Taiwan's construction sector, and IT experience in software design. The efficiency improvement targets will be confirmed in a pilot with 1–2 contractors before any commitment to customers.

8. Roadmap & Use of Funds

Funding would go to the MVP, a real pilot, and network growth.

- **Phase 1 — SiteProof MVP:** Build the five must-have functions (Order → Delivery → Inspection → Acceptance → Completion, where completion includes retention and the supplier rating); pilot with 1–2 contractors to validate impact; storage for quality and ESG documents.
- **Phase 2 — SiteProof Expansion:** RFQ & quotation comparison; LINE notifications and digital payment requests; advanced analytics, Pro-tier features, featured listings and mobile optimization.
- **Phase 3 — SiteProof Intelligence:** OCR for quality inspection PDFs; budget-overrun alerts; GPS / logistics and ERP integration; predictive delivery-risk analytics.

ANNEX: https://chatgpt.com/share/6abcb1cc-4cd0-83ee-be31-543572b67b7d